STAT 576: Empirical Process Theory and Weak Convergence Lecture Notes

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1 Statistical–Computational Trade-offs

A recurring theme in high-dimensional probability, random optimization, and statistical inference is the gap between what is information-theoretically possible and what is computationally feasible.

Problem 1 (Existence)**.** When does a desirable object exist with high probability?

Problem 2 (Efficient search)**.** When can an efficient algorithm find such an object?

The first question is usually called the *statistical threshold* or *information-theoretic threshold*. The second question concerns the *algorithmic threshold* or *computational threshold*.

Definition 1 (Statistical–Computational Gap)**.** A random model is said to exhibit a statistical–computational gap if there exists a range of parameters for which the target object exists with high probability, but no known polynomial-time algorithm can find it.

Remark. The point is not that polynomial-time algorithms are formally impossible in every such regime. Rather, the model displays a parameter range where existence can be proved nonconstructively, while efficient search appears to fail.

2 Maximum Independent Set in $G(n, 1/2)$

Let $G \sim G(n, 1/2)$. An *independent set* is a set of vertices containing no edges between any two of its vertices.

Problem 3. Given $G \sim G(n, 1/2)$, find a large independent set.

Let $\alpha(G)$ denote the size of the maximum independent set of G.

Theorem 1 (Matula, 1976)**.** *For $G \sim G(n, 1/2)$,*

$$\alpha(G) = 2(1 + o(1)) \log_2 n$$

with high probability as $n \to \infty$.

This is an information-theoretic statement: it says that independent sets of this size exist, but it does not by itself give an efficient algorithm for finding one.

Example 1 (Greedy algorithm)**.** Label the vertices as $v_1, \dots, v_n$. Define $I_0 = \varnothing$. At step t, set

$$I_t = \begin{cases} I_{t-1} \cup \{v_t\}, & \text{if } v_t \text{ has no edge to any vertex in } I_{t-1}, \\ I_{t-1}, & \text{otherwise.} \end{cases}$$

This algorithm returns an independent set, but only of size roughly $\log_2 n$, far below the optimal scale $2 \log_2 n$.

Open Problem 1 (Karp, 1976)**.** *For a fixed $\varepsilon > 0$, find a polynomial-time algorithm returning an independent set of size*

$$(1 + \varepsilon) \log_2 n,$$

or prove that doing so is computationally hard.

Remark. The gap between $\log_2 n$ and $2 \log_2 n$ is a prototypical statistical–computational trade-off: the optimal object exists, but known efficient algorithms do not reach the information-theoretic threshold.

3 Sparse Regression

Consider the linear model

$$Y = X\beta^* + W,$$

where

$$Y \in \mathbb{R}^n, \qquad X \in \mathbb{R}^{n \times p}, \qquad W \in \mathbb{R}^n.$$

Assume the rows of X are sub-Gaussian and W has independent noise entries. The parameter β^* is assumed to be sparse:

$$\|\beta^*\|_0 \le s.$$

Problem 4. Recover the support or the vector β^* from (X, Y).

A standard sample-size scale for sparse recovery is

$$n \gtrsim s \log(p/s).$$

Theorem 2 (Sparse recovery threshold, informal)**.** *Under suitable design conditions and signal-strength assumptions, ℓ_1-based methods can recover sparse signals when*

$$n \gtrsim s \log(p/s).$$

Remark. The quantity $s \log(p/s)$ is the effective dimension of the sparse model class. It appears both in information-theoretic and algorithmic analyses of sparse recovery.

Theorem 3 (Wainwright-type threshold; to verify)**.** *There is an information-theoretic threshold for sparse recovery depending on s, p, the noise level, and the signal strength. A typical form is*

$$n \gtrsim \frac{s \log(p/s)}{\log(1 + \mathrm{SNR})},$$

up to constants and model-specific assumptions.

Remark. The exact expression depends on whether the goal is exact support recovery, approximate recovery, or estimation in norm. This statement should be checked against the original lecture source before being quoted as a formal theorem.

4 Planted Clique

Let $G(n, 1/2)$ denote the Erdős–Rényi random graph on n vertices, where each edge is independently present with probability $1/2$.

Definition 2 (Planted Clique Model)**.** Fix $k \in \mathbb{N}$. The planted clique model PC_k is generated as follows:

1. Sample $G \sim G(n, 1/2)$.
2. Choose k vertices uniformly at random.
3. Add all missing edges among these k vertices.

The selected k vertices form a planted clique.

There are two natural problems.

Problem 5 (Detection)**.** Given a graph G, decide between

$$H_0 : G \sim G(n, 1/2)$$

and

$$H_1 : G \sim PC_k.$$

Problem 6 (Recovery)**.** Given $G \sim PC_k$, recover the planted clique.

The information-theoretic threshold for detection is around

$$k \asymp 2 \log_2 n.$$

Indeed, the largest clique in $G(n, 1/2)$ has size approximately $2 \log_2 n$, so planted cliques below this scale are statistically hidden among naturally occurring cliques.

Known polynomial-time algorithms succeed when

$$k \gtrsim \sqrt{n},$$

up to logarithmic factors.

Remark. If

$$k = \Omega(\sqrt{n}\log^c n),$$

then one can show that the k vertices with largest degrees must come from the planted clique. This gives a simple degree-based recovery method in a sufficiently large-signal regime.

Conjecture 1 (Planted Clique Conjecture)**.** *Suppose*

$$\limsup_{n\to\infty} \frac{\log k}{\log n} < \frac{1}{2}.$$

For example, $k = n^{1/2-\varepsilon}$. Then no polynomial-time test is believed to reliably distinguish $G(n, 1/2)$ from PC_k.

Equivalently, for any computationally efficient test

$$\phi : \mathcal{G}_n \to \{0, 1\},$$

one expects

$$\mathbb{P}_{H_0}\{\phi(G) = 1\} + \mathbb{P}_{H_1}\{\phi(G) = 0\}$$

to stay bounded away from 0*.*

Remark. A fully formal version of the planted clique conjecture must specify the class of algorithms being ruled out, typically randomized polynomial-time algorithms. The above statement is the standard informal statistical formulation used as a hardness assumption.

Remark. The planted clique conjecture implies computational lower bounds for several problems, including variants of sparse PCA and related high-dimensional inference tasks.

5 Overlap Gap Property

The overlap gap property, abbreviated OGP, is a geometric explanation for algorithmic failure in certain random optimization problems.

Let S be the solution space of a random optimization problem. For two solutions $\sigma, \tau \in S$, define their normalized overlap by

$$R(\sigma, \tau) = \frac{1}{n} \langle \sigma, \tau \rangle .$$

Equivalently, one may measure their normalized Hamming distance:

$$\frac{1}{n} d_H(\sigma, \tau).$$

Definition 3 (Overlap Gap Property, informal)**.** A random model exhibits the overlap gap property if, with high probability, there exist constants

$$0 \le \nu_1 < \nu_2 \le 1$$

such that for all pairs of solutions $\sigma, \tau \in S$,

$$\frac{1}{n} d_H(\sigma, \tau) \in [0, \nu_1] \cup [\nu_2, 1].$$

Equivalently, there is a forbidden interval of intermediate distances.

Intuition. The solution space splits into clusters. Two solutions are either very close or very far apart, but not at an intermediate distance. This fragmented geometry can obstruct stable local or incremental algorithms.

Theorem 4 (Informal OGP-to-hardness principle)**.** *If a model exhibits OGP, then large classes of stable algorithms fail to find solutions in the OGP regime.*

Remark. The precise algorithm class depends on the model. Typical examples include local algorithms, low-degree polynomial algorithms, Markov chain methods, and certain classes of incremental algorithms.

6 Binary Symmetric Perceptron

The binary symmetric perceptron is a random constraint satisfaction problem originally introduced in the study of limits of Glauber dynamics.

Let

$$X_1, \dots, X_m \overset{iid}{\sim} N(0, I_n), \qquad m = \alpha n.$$

Fix $K > 0$. Define the solution space

$$S_\alpha(K) = \left\{ \sigma \in \{-1,1\}^n : \frac{1}{\sqrt{n}} |\langle X_i, \sigma \rangle| \le K \text{ for all } i = 1, \dots, m \right\}.$$

The parameter $\alpha = m/n$ is the constraint density or aspect ratio.

Problem 7 (Satisfiability)**.** For which α is

$$S_\alpha(K) \neq \varnothing$$

with high probability?

Problem 8 (Efficient search)**.** For which α can an efficient algorithm find an element of $S_\alpha(K)$?

For a fixed $\sigma \in \{-1,1\}^n$,

$$\frac{1}{\sqrt{n}} \langle X_i, \sigma \rangle \sim N(0,1).$$

Therefore

$$\mathbb{P}\left\{ \frac{1}{\sqrt{n}} |\langle X_i, \sigma \rangle| \le K \right\} = \mathbb{P}\{|N(0,1)| \le K\}.$$

Since there are 2^n possible assignments σ, the expected number of solutions is

$$\mathbb{E}\,|S_\alpha(K)| = 2^n \mathbb{P}\{|N(0,1)| \le K\}^{\alpha n}.$$

Thus the first-moment threshold is

$$\alpha_c(K) = -\frac{1}{\log_2 \mathbb{P}\{|N(0,1)| \le K\}}.$$

If K is small, then

$$\mathbb{P}\{|N(0,1)| \le K\} \asymp K,$$

and hence

$$\alpha_c(K) \asymp \frac{1}{\log(1/K)}.$$

Theorem 5 (Satisfiability threshold; to verify)**.** *Let $K > 0$ be fixed. Then*

$$\lim_{n\to\infty} \mathbb{P}\{S_\alpha(K) \neq \varnothing\} = \begin{cases} 1, & \alpha < \alpha_c(K), \\ 0, & \alpha > \alpha_c(K). \end{cases}$$

Remark. This theorem gives the information-theoretic threshold. It does not imply the existence of an efficient algorithm up to $\alpha_c(K)$.

Known algorithms only work up to a smaller constraint density. The board indicates an algorithmic regime of the form

$$\alpha \lesssim K^2,$$

or a nearby expression involving logarithmic factors.

Remark. The exact algorithmic threshold should be verified from the lecture source before quoting. The photographed board is not fully legible at this point.

The resulting picture is

$$\text{Algorithmic threshold} \quad < \quad \text{OGP threshold} \quad < \quad \text{Satisfiability threshold.}$$

Thus the binary symmetric perceptron provides another example of a statistical–computational gap.

Summary

The lecture develops several examples where the existence threshold and the efficient-algorithm threshold differ.

1. In maximum independent set, the largest independent set has size about $2\log_2 n$, but simple algorithms find only about $\log_2 n$.
2. In sparse regression, sample complexity depends on the effective sparse dimension $s\log(p/s)$, with thresholds varying according to signal strength and recovery criterion.
3. In planted clique, statistical detection is possible near $2\log_2 n$, but polynomial-time algorithms are only known near $\sqrt{n}$.
4. In the binary symmetric perceptron, the satisfiability threshold is governed by $\alpha_c(K)$, while known efficient algorithms work below the information-theoretic threshold.
5. The overlap gap property gives a geometric explanation for why efficient algorithms may fail before the statistical threshold.

Items to Verify

1. The exact Wainwright sparse-regression threshold and its assumptions.
2. The exact algorithmic threshold for binary symmetric perceptron.
3. The lecture number, date, and scriber information.